

PROJECT TITLE:
R.O.A.D.S (REAL-TIME ONBOARD ANOMALY DETECTION SYSTEM)

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KOLEJ POLY-TECH MARA ALOR SETAR

2026



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DIPLOMA IN COMPUTER SCIENCE

TTS3424

IT PROJECT

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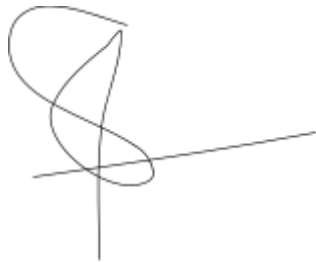
MADAM NUR NADIRAH BINTI HARUN

DECLARATION

Name : WAN ZAHIN SYAFIQHAH BINTI ABDUL FATTAH

I.D. Number : ASJ243510368

I hereby declare that this project paper is the result of my own work, except for quotations and summaries that duly acknowledged.

A handwritten signature in black ink, consisting of a stylized 'S' shape with a horizontal line extending to the right.

Signature:

Date: 17/3/2026

ACKNOWLEDGEMENT

Alhamdulillah, first and foremost, I would like to express my deepest gratitude to Allah S.W.T for giving me the strength, patience, and ability to complete this project successfully.

I would like to extend my sincerest appreciation to my supervisor, Madam Nur Nadirah Binti Harun, for her invaluable guidance, continuous support, constructive feedback, and encouragement throughout the entire development process of this project. Her expertise and dedication have been instrumental in shaping the direction and quality of this work.

I would also like to thank the Faculty of Computer Science and Information Technology at Kolej Poly-Tech MARA Alor Setar for providing the necessary facilities, equipment, and technical resources to successfully develop the R.O.A.D.S prototype.

On a personal level, I must acknowledge the profound impact that the world of Formula 1 has had on my journey. Watching the relentless pursuit of perfection, the technical precision, and the "never give up" mentality of the sport provided me with the mental clarity and motivation needed to start and complete this project. The resilience shown on the track served as a constant reminder to stay focused under pressure and to keep pushing for the best possible results.

Special thanks are extended to my family for their unwavering moral support and encouragement, and to my friends for their cooperation and assistance throughout this journey.

Finally, I would like to thank all parties who have directly or indirectly contributed to the completion of this project. Your support is deeply appreciated — simply lovely.

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CHAPTER 1: INTRODUCTION AND BACKGROUND STUDY

1.1 Introduction of Project

Road infrastructure maintenance is one of the most persistent challenges in urban and rural transportation planning, particularly in tropical countries such as Malaysia where frequent heavy rainfall accelerates the deterioration of asphalt road surfaces. The formation of potholes and road surface anomalies poses significant risks to motorist safety, causes avoidable vehicle damage, and contributes to traffic accidents that could otherwise be prevented with timely detection and warning.

R.O.A.D.S, which stands for Real-Time Onboard Anomaly Detection System, is a low-cost embedded IoT system designed to address this problem by transforming standard vehicles into intelligent mobile road condition monitoring nodes. The system utilises a Cytron Maker Feather AIoT S3 microcontroller, built on the ESP32-S3 chip, interfaced with an MPU6050 6-axis Inertial Measurement Unit (IMU) sensor mounted on the vehicle chassis. Vibration data captured at 77Hz is classified in real time on the device using an Edge Artificial Intelligence model trained via the Edge Impulse platform.

The classified road condition data is then transmitted wirelessly via Bluetooth Low Energy (BLE) to a custom-developed Android mobile application. Upon detecting a high-confidence pothole event, the application displays a real-time visual warning on an animated road interface and simultaneously logs the event to a Firebase Realtime Database, creating a crowd-sourced record of road hazards accessible to authorities and fleet managers.

This project demonstrates the practical application of embedded systems, Edge AI, wireless communication, mobile application development, and cloud computing to solve a real-world infrastructure problem in a cost-effective and scalable manner.

1.2 Problem Statements

Current road hazard reporting systems rely heavily on manual processes such as driver-initiated reports through navigation applications or scheduled inspections by municipal authorities. These approaches have three critical limitations.

First, they are inherently reactive rather than proactive. Hazards are only reported after an accident or vehicle damage has already occurred, rather than warning drivers before they encounter the danger. There is no automated mechanism to detect and communicate hazards to drivers in real time.

Second, existing manual reporting systems are highly delayed. The time between a pothole forming, a driver reporting it, and a maintenance crew addressing it can span weeks or months. During this period, the hazard continues to cause vehicle damage and risk driver safety.

Third, traditional inspection methods lack real-time localised data. Without continuous monitoring, authorities cannot prioritise repairs based on actual hazard frequency and severity. Decisions are made reactively based on complaints rather than data-driven insights.

There is therefore a critical need for an automated, low-cost system that can detect road anomalies instantly at the vehicle level, provide immediate driver warnings, and log hazard data to the cloud for infrastructure maintenance planning.

1.3 Project Objectives

The following objectives were established to guide the development of R.O.A.D.S:

1. To design and implement an IoT hardware node using the Cytron Maker Feather AIoT S3 (ESP32-S3) microcontroller and MPU6050 6-axis for capturing chassis vibration data at a sampling frequency of 77Hz.
2. To train and deploy an Edge AI classification model via Edge Impulse capable of classifying road conditions into three categories — Pothole, Bump, and Smooth Road — with a minimum inference accuracy of 80%.
3. To develop a custom Android mobile application that receives classified BLE telemetry from the hardware node and displays real-time road condition visualisation on an animated road interface.
4. To integrate Firebase Realtime Database for automatic cloud logging of all detected pothole events, including severity score, timestamp, and repair status.

5. To evaluate the overall system performance in terms of classification accuracy, BLE transmission latency, and end-to-end response time from detection to mobile notification.

1.4 Project Scope and Limitations

The scope of this project covers the complete development pipeline from hardware sensor integration to mobile application delivery. This includes firmware development on the ESP32-S3, Edge Impulse model training and on-device deployment, BLE communication between the hardware node and an Android device, real-time road condition visualisation through a custom Canvas-rendered Android interface, and Firebase cloud integration for persistent hazard logging.

The system is developed and tested using a model vehicle chassis on a simulated road surface within a controlled laboratory environment. This approach allows consistent and repeatable testing conditions for model validation and system performance evaluation.

The following limitations apply to the current prototype. First, the system does not include a dedicated GPS hardware module. Precise geographic coordinates are not appended to BLE payloads in the current implementation and are identified as future work. Second, the Edge AI model was trained on data collected from a single prototype vehicle chassis and may require retraining for accurate classification across different vehicle types and road surfaces. Third, the BLE communication range is limited to approximately 10 metres, requiring the mobile device to remain in proximity to the hardware node during operation. Fourth, the system requires an active internet connection on the mobile device for Firebase synchronisation, though BLE detection and local display function without connectivity.

CHAPTER 2: PLANNING AND ANALYSIS

2.1 Project Planning – Gantt Chart

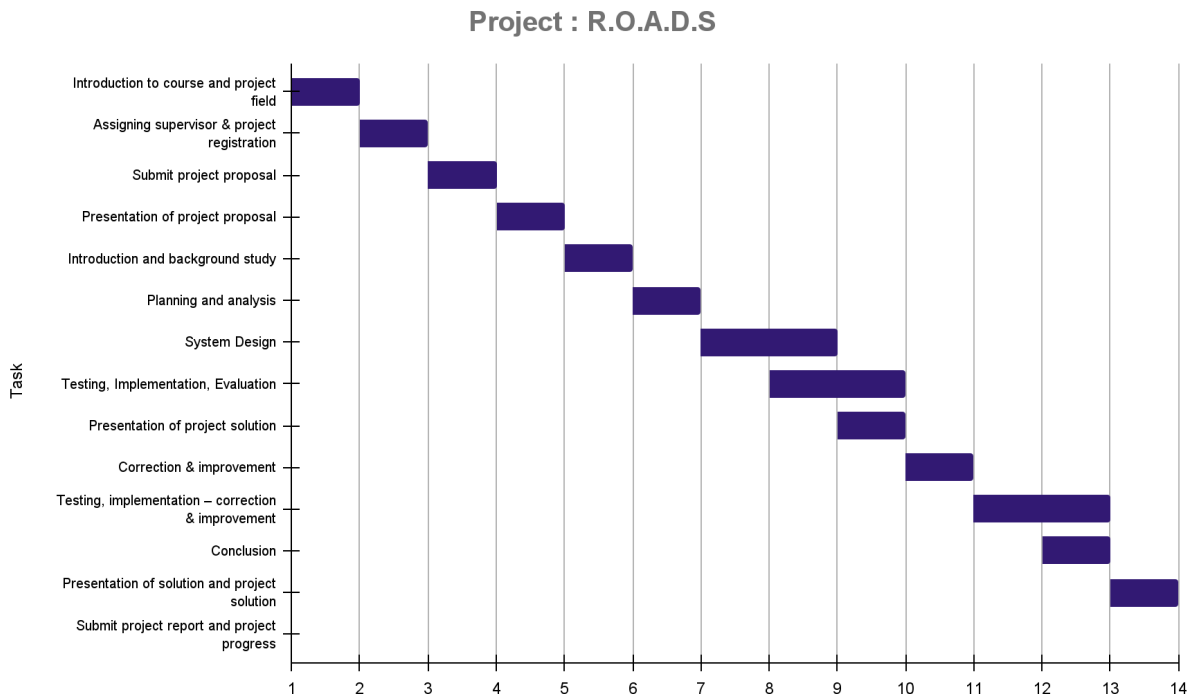


Table 2.1: Project Gantt Chart

1) PLANNING

Week 1: Conducted research to find a suitable title for the project. Existing road monitoring systems were reviewed to identify gaps and opportunities for improvement.

Week 2: A supervisor was assigned and a discussion was held to present the proposed project title. Feedback was used to confirm the project direction and technical feasibility.

Week 3: The project proposal was completed and submitted to the supervisor, outlining the problem statement, objectives, scope, and methodology.

2) ANALYSIS

Week 4: The project proposal presentation was conducted. Feedback from the supervisor was noted and incorporated into the project plan.

Week 5: The introduction section of the project report was drafted and a comprehensive background study was conducted, reviewing related work on road monitoring systems, Edge AI, BLE communication, and Firebase integration.

Week 6: A detailed project plan was developed outlining all tasks, timelines, and resources. Requirement analysis was conducted through survey and observation methods to gather user needs and validate the proposed system.

3) DESIGN

Week 7 to Week 8: Initial system design and prototypes were developed, covering hardware wiring, firmware architecture, Edge AI model training using Edge Impulse, and Android application interface design. Design plans were reviewed for feasibility and alignment with project objectives.

Week 9: Component testing was conducted to identify any discrepancies and ensure functionality and reliability of the hardware sensor, BLE transmission, Edge AI model, and Firebase operations.

Week 10: The project solution presentation was conducted to present the current system prototype to the supervisor and receive feedback.

Week 11: Areas requiring correction were identified based on feedback and testing results. Necessary improvements were implemented including BLE stability fixes, animation improvements, and Firebase synchronisation refinements.

4) IMPLEMENTATION

Week 12 to Week 13: Full system integration testing was executed, covering end-to-end latency, BLE range, Firebase synchronisation, and user acceptance testing. Corrective measures were applied based on testing outcomes and project findings were summarised and documented.

Week 14: The final project presentation was conducted and the completed project report was submitted.

2.2 Requirement Analysis (Data Gathering)

1. What is your age range?

19 responses

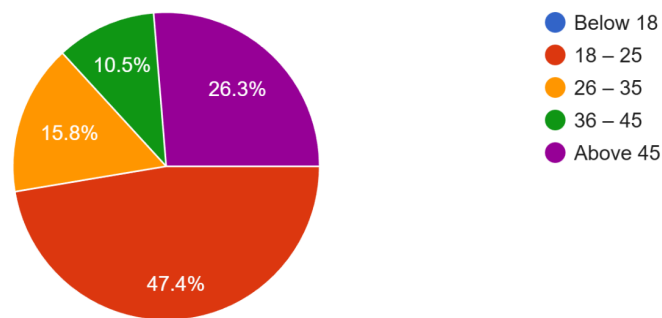


Figure 2.2.1: Age Range of Respondents

2. What is your occupation?

19 responses

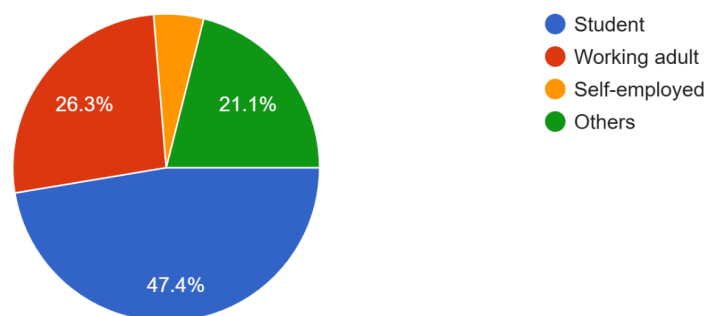


Figure 2.2.2: Occupation of Respondents

3. How often do you travel by road?

19 responses

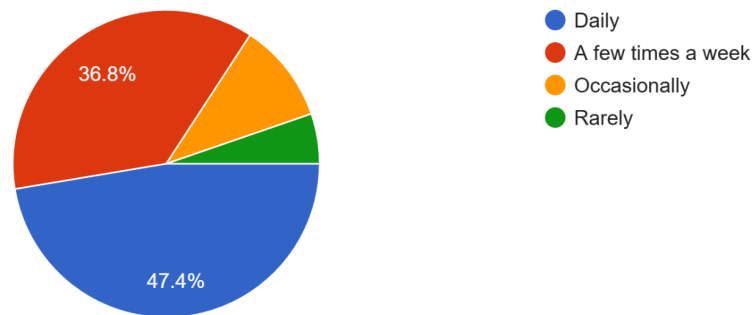


Figure 2.2.3: Frequency of Road Travel

4. What type of vehicle do you usually use?

19 responses

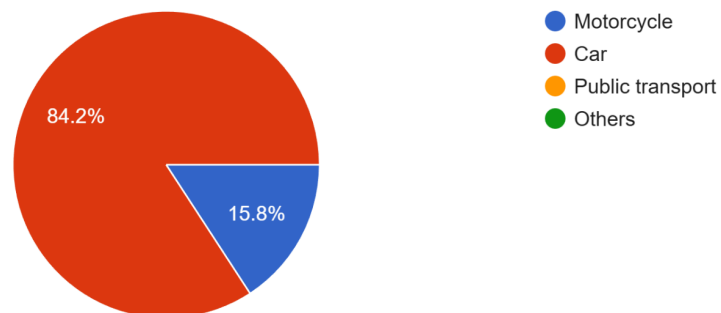


Figure 2.2.4: Type of Vehicle Used

5. How often do you encounter potholes or road damage during your daily travel?

19 responses

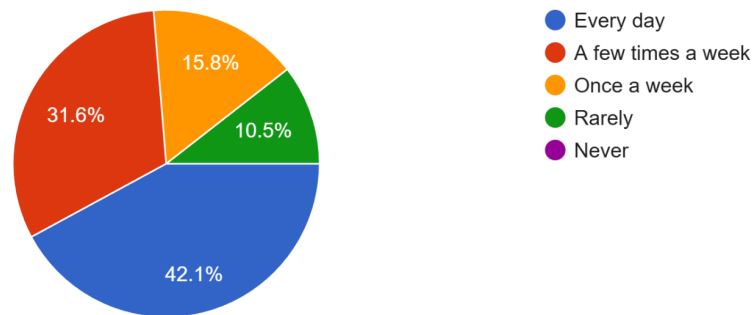


Figure 2.2.5: Frequency of Encountering Potholes or Road Damage

6. Have you ever experienced vehicle damage caused by potholes or road anomalies? (e.g. tyre puncture, suspension damage)

19 responses

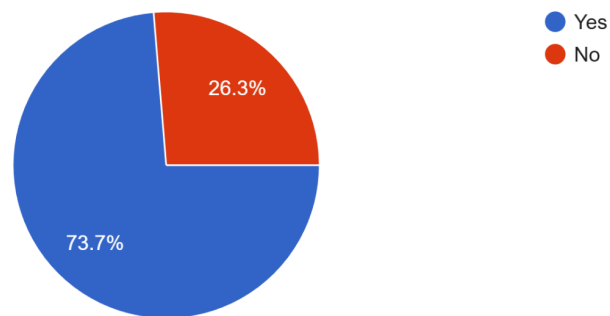


Figure 2.2.6: Experience of Vehicle Damage Caused by Potholes

7. Have you ever been involved in or witnessed an accident caused by a pothole or road damage?

19 responses

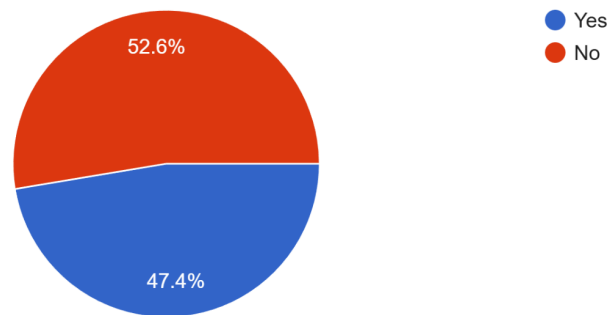


Figure 2.2.7: Involvement or Witnessing of Accident Caused by Road Damage

8. How would you rate the overall road condition in your area?

19 responses

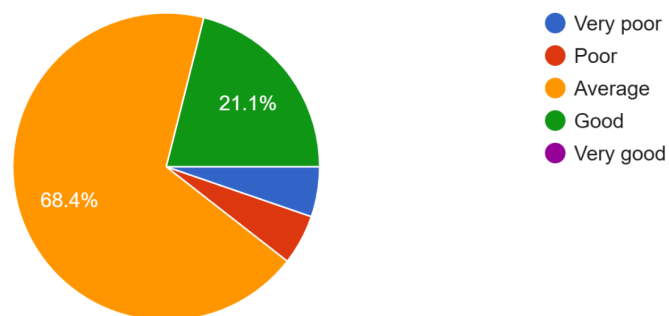


Figure 2.2.8: Rating of Overall Road Condition

9. Have you ever reported a road damage or pothole to the authorities?

19 responses

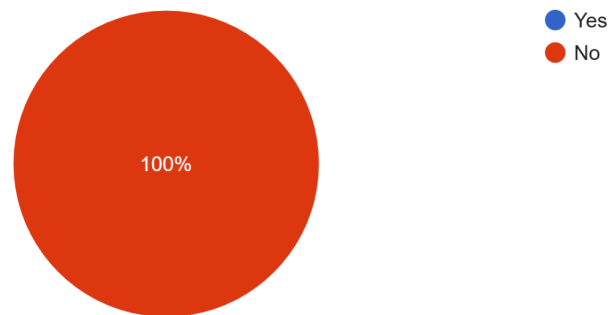


Figure 2.2.9: Experience of Reporting Road Damage to Authorities

9. Have you ever reported a road damage or pothole to the authorities?

19 responses

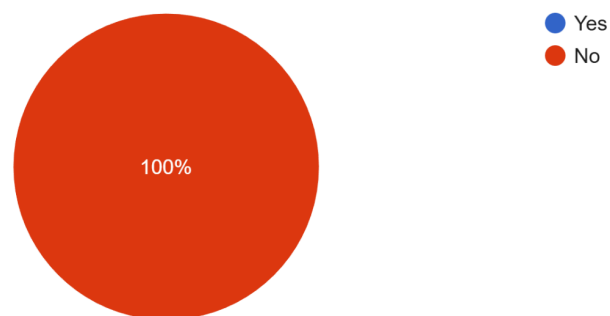


Figure 2.2.10: Method Used to Report Road Damage

11. Are you aware of any existing automated road condition monitoring system?

19 responses

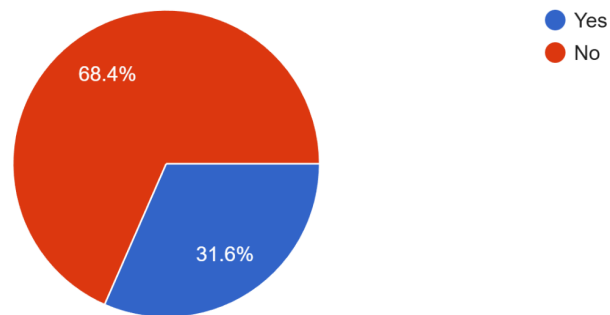


Figure 2.2.11: Awareness of Existing Automated Road Condition Monitoring System

12. Have you ever used any application to report road damage?

19 responses

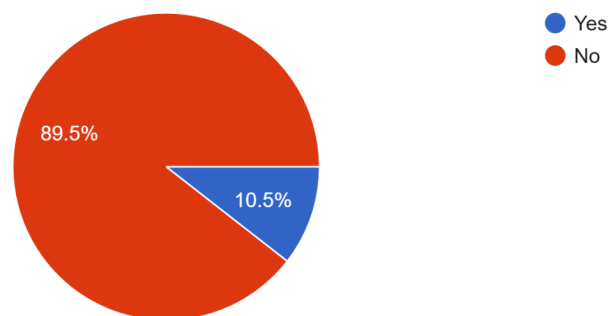


Figure 2.2.12: Experience of Using Application to Report Road Damage

13. If yes, were you satisfied with the response from the authorities after reporting?

19 responses

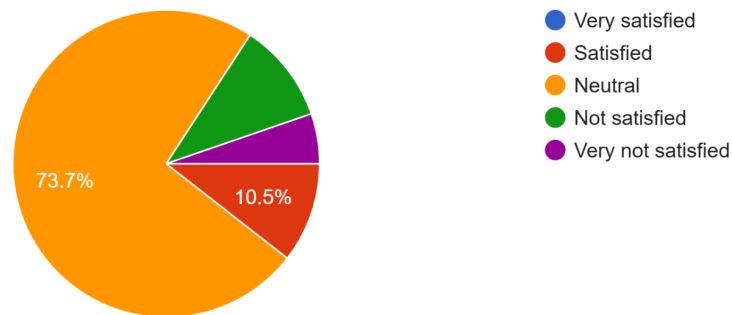


Figure 2.2.13: Satisfaction with Authority Response After Reporting

14. Do you think there is a need for an automated system that can detect and warn drivers about potholes in real time?

19 responses

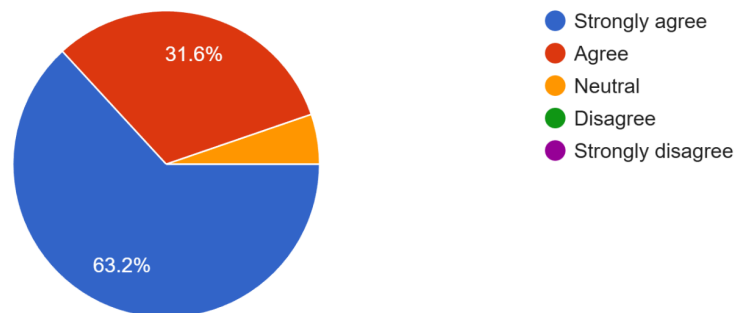


Figure 2.2.14: Need for Automated Real-Time Pothole Detection System

15. Would you be interested in using a system that automatically detects potholes and displays a warning on your phone while driving?

19 responses

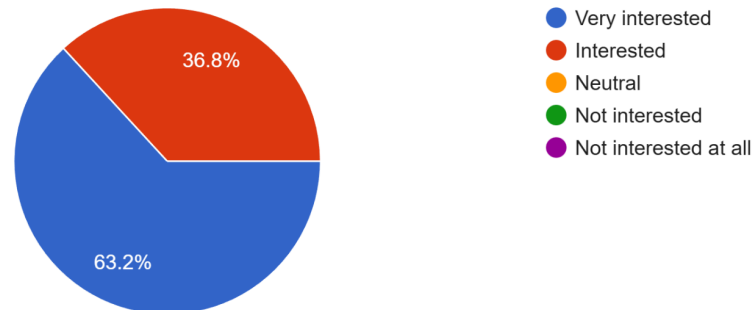


Figure 2.2.15: Interest in Using Automated Pothole Warning System

16. Which feature do you think is most important in a road monitoring system? (Choose one)

19 responses

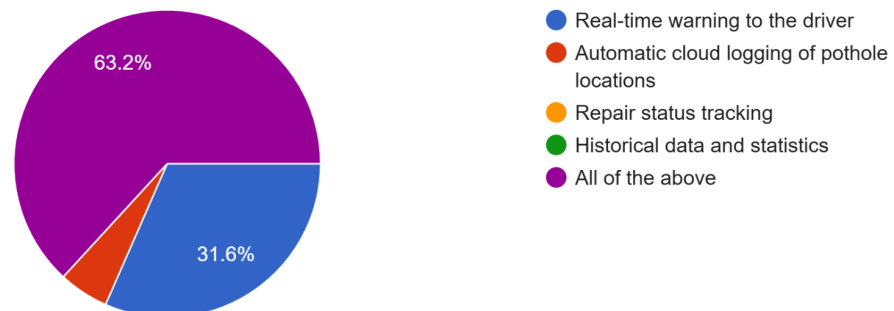


Figure 2.2.16: Most Important Feature in a Road Monitoring System

17. How important is it for road hazard data to be shared with local authorities for maintenance purposes?

19 responses

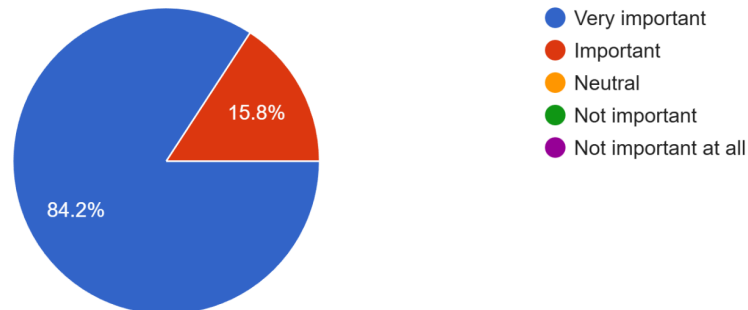


Figure 2.2.17: Importance of Sharing Road Hazard Data with Authorities

18. What device would you prefer to receive road condition warnings on?

19 responses

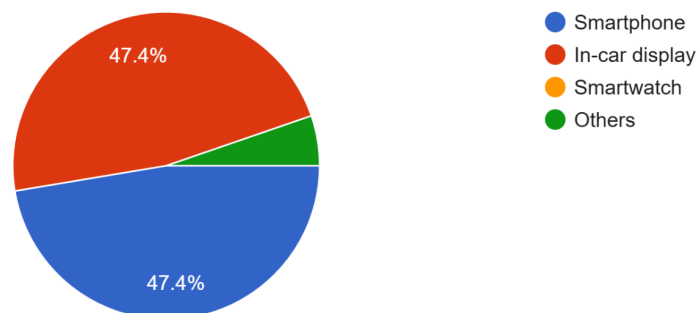


Figure 2.2.18: Preferred Device to Receive Road Condition Warnings

19. Would you be willing to have this system installed in your vehicle?

19 responses

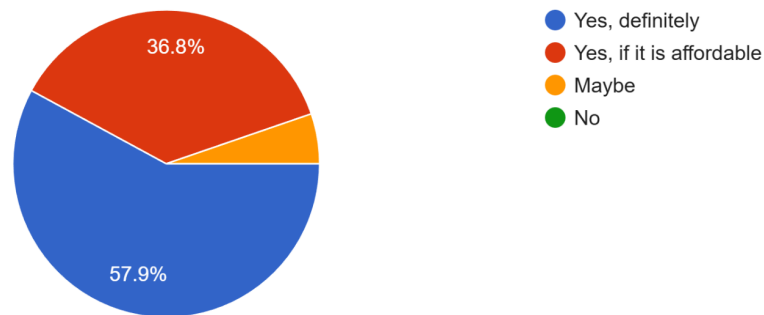


Figure 2.2.19: Willingness to Have System Installed in Vehicle

20. Any suggestions or comments regarding road condition monitoring systems?

6 responses

no

The system should be affordable, accurate, and able to provide real-time road condition alerts.

nope

just focus on your journey, please concentrate and dont play phone

This shall be implemented all over Malaysia, the sooner the better. Bet it's gonna be live and cost saving for most of the road users!

Figure 2.2.20: Suggestions and Comments from Respondents

2.3 System Analysis

2.3.1 Use Case Diagram

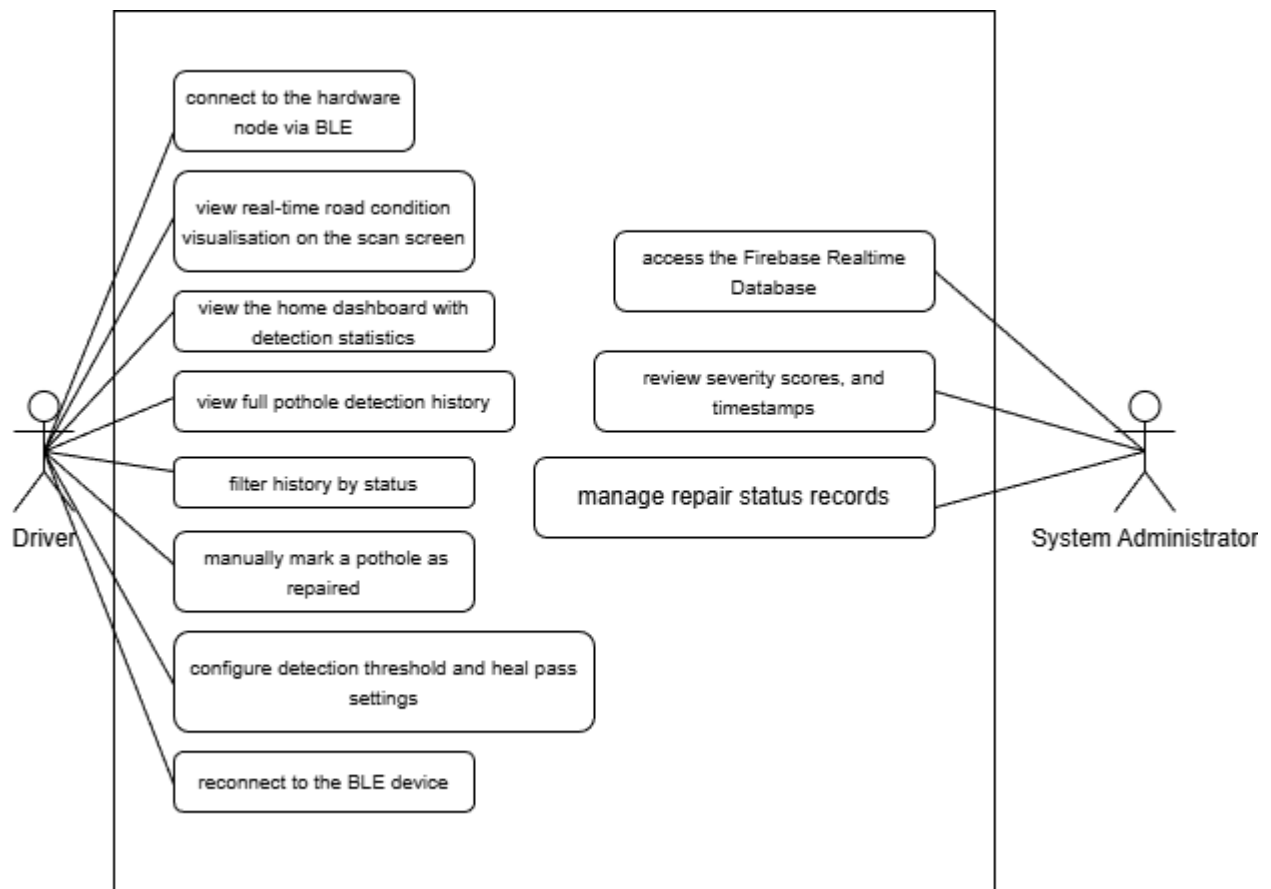


Figure 2.3.1: Use Case Diagram of R.O.A.D.S System

2.3.2 Activity Diagram

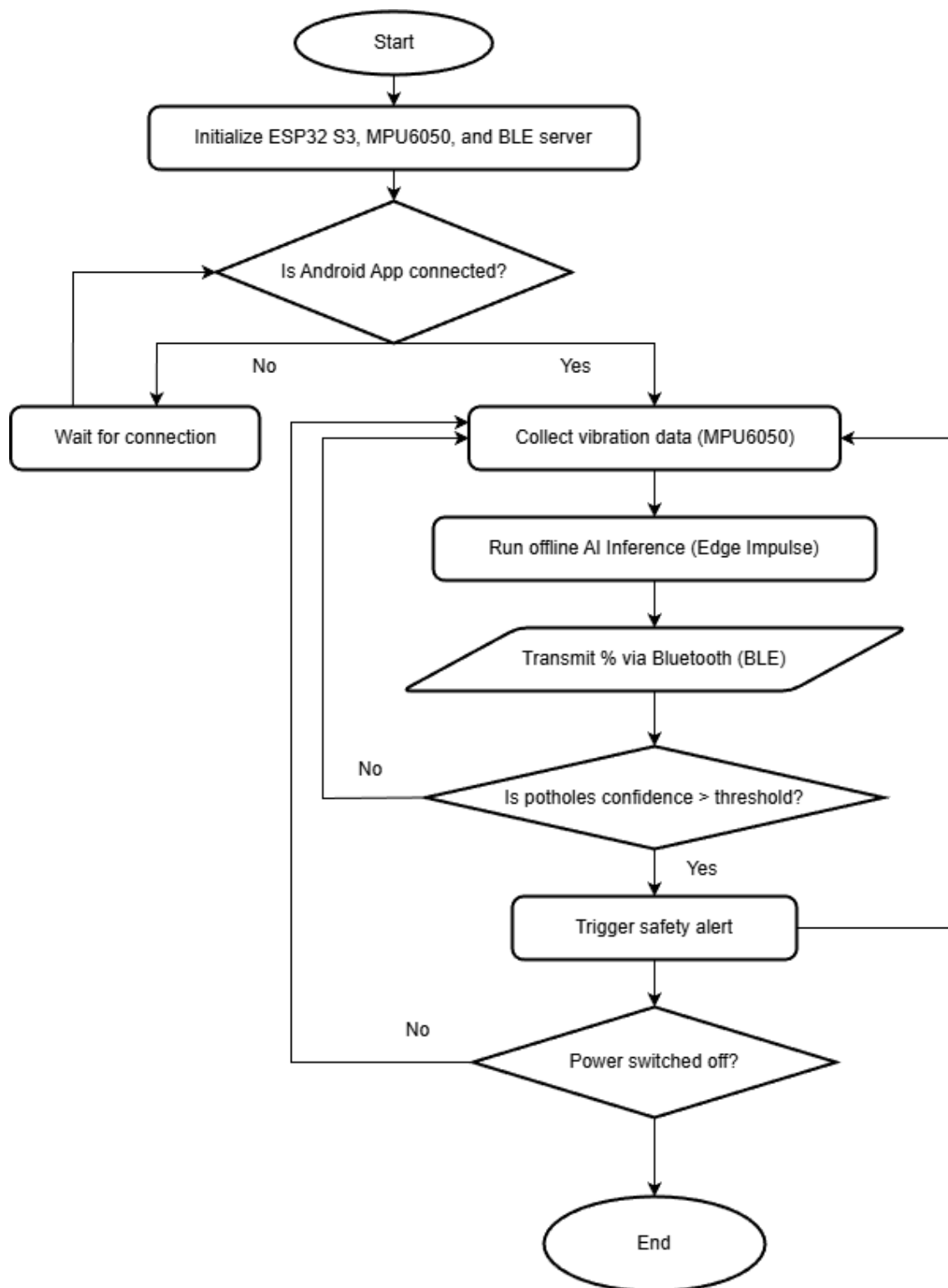


Figure 2.3.2: Activity Diagram of R.O.A.D.S System

2.3.3 Class Diagram

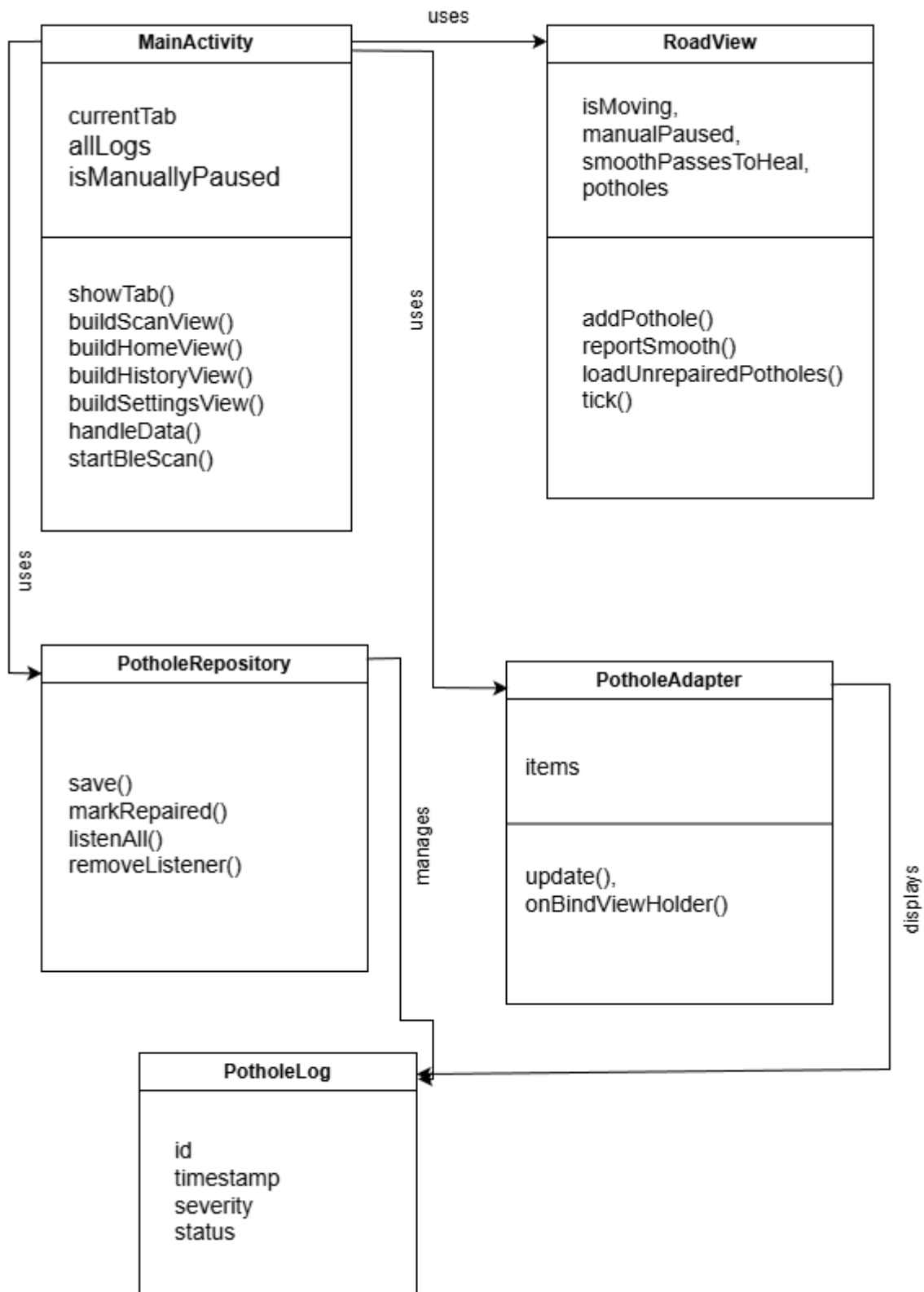


Figure 2.3.3: Class Diagram of R.O.A.D.S System

CHAPTER 3: DESIGN

3.1 Software and Hardware Requirement

Software Requirements

Software	Version	Purpose
Arduino IDE	2.x	Firmware development and flashing to ESP32-S3
Edge Impulse Studio	Web-based	Dataset collection, spectral feature extraction, neural network training, and C++ model export
Android Studio	Flamingo or above	Android application development in Kotlin
Firebase Console	Web-based	Cloud database management and real-time data monitoring
Git	Latest	Version control for firmware and application source code
Draw.io	Web-based	System architecture and UML diagram creation
Google Docs	Web-based	Project report documentation

Hardware Requirements

Hardware	Specification	Purpose
Cytron Maker Feather AloT S3	ESP32-S3 chip, 8MB Flash, 2MB PSRAM	Main microcontroller for sensor reading, Edge AI inference, and BLE transmission
MPU6050	6-axis accelerometer and gyroscope, I2C interface	Captures chassis vibration data at 77Hz for road condition classification
Prototype Vehicle Chassis	Model car chassis with motor	Physical platform for sensor mounting and road surface simulation
Android Smartphone	Android 8.0 (API 26) or above, BLE capable	Runs the companion mobile application and connects to hardware node via BLE
Laptop	Windows 10 or above	Used for firmware development, model training, and application development
USB Cable	USB-C	Firmware flashing and serial debugging

3.2 Interface Design

Screen 1: Home Screen

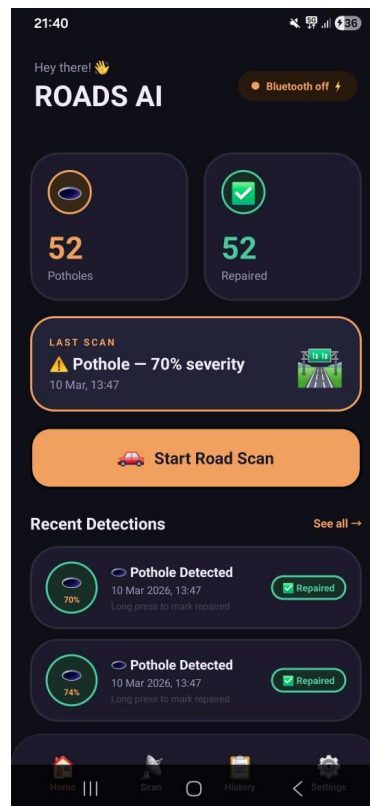


Figure 3.2.1: Home Screen

Figure 3.2.1 shows the Home Screen of the R.O.A.D.S Android application. The home screen serves as the main dashboard of the system. At the top of the screen, the application title ROADS AI is displayed alongside a Bluetooth connection status badge that shows the current BLE connection state. When Bluetooth is off, the badge displays "Bluetooth off" and when connected it displays the connection status with a colour-coded indicator.

Below the header, two statistic cards are displayed side by side showing the total number of potholes detected and the total number of potholes marked as repaired, both retrieved in real time from Firebase Realtime Database. A Last Scan card below the statistics displays the most recent pothole detection event including the severity percentage and timestamp. A large orange Start Road Scan button is provided for the user to navigate directly to the Scan screen. At the bottom of the screen, a

Recent Detections section displays the latest pothole detection records from Firebase with severity percentage bubbles, detection titles, timestamps, and repair status pills.

Screen 2: Scan Screen

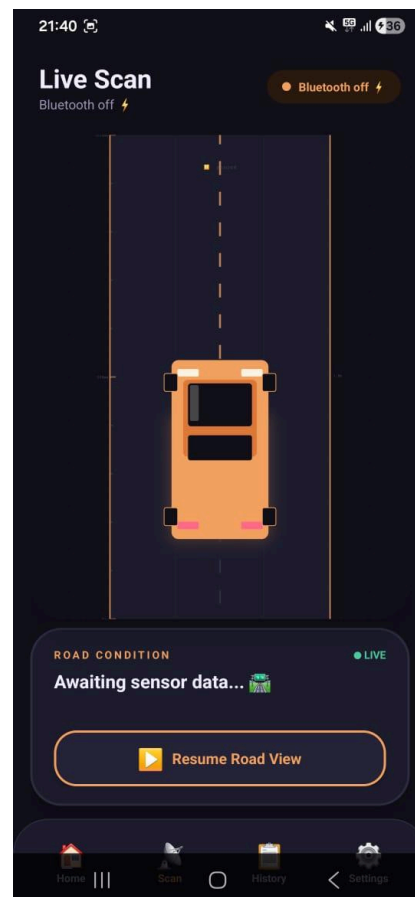


Figure 3.2.2: Scan Screen

Figure 3.2.2 shows the Scan Screen of the R.O.A.D.S Android application. The scan screen is the primary real-time road monitoring interface of the system. At the top of the screen, the title Live Scan is displayed alongside the current Bluetooth connection status pill. The main area of the screen is occupied by a custom Canvas-rendered animated road view that displays a dark asphalt road with orange curb lines and a peach-coloured car positioned at the centre of the road. The road scrolls continuously to simulate vehicle movement. When a pothole is detected via BLE, an animated warning triangle sign appears on the road with a bounce-in animation and pulsing radar rings.

At the bottom of the screen, a Road Condition label displays the latest classification result received from the hardware node. A Pause or Resume Road View button is provided to allow the user to manually freeze or resume the road animation. When the road view is paused, new pothole detection from BLE is also suspended.

Screen 3: History Screen

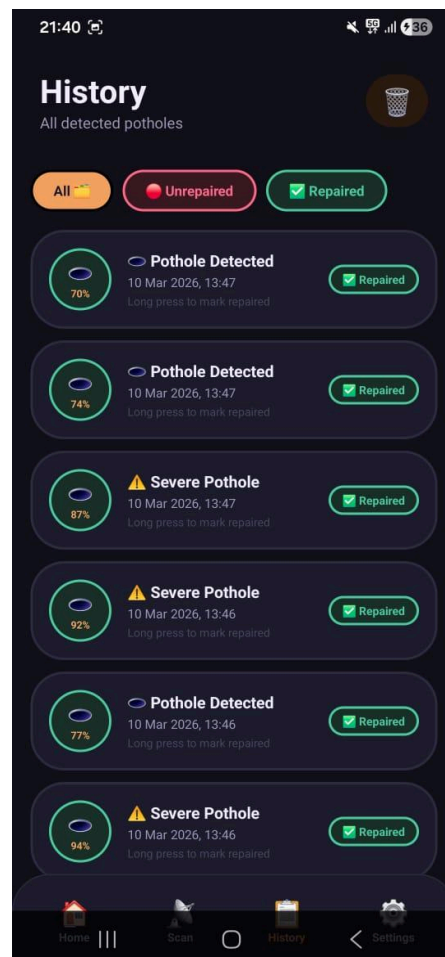


Figure 3.2.3: History Screen

Figure 3.2.3 shows the History Screen of the R.O.A.D.S Android application. The history screen displays a complete log of all pothole detection records stored in Firebase Realtime Database. At the top of the screen, the title History is displayed alongside a delete button for clearing all records from the database. Below the header, three filter chip buttons labelled All, Unrepaired, and Repaired allow the user to filter the displayed records by repair status.

Each list item displays a severity bubble showing the pothole percentage score, a detection title indicating the severity level such as Pothole Detected or Severe Pothole, the detection timestamp, and a status pill colour-coded in green for repaired and red for unrepaired. Users can long-press any unrepaired item to manually mark it as repaired, which updates the record in Firebase and refreshes the list automatically.

Screen 4: Settings Screen

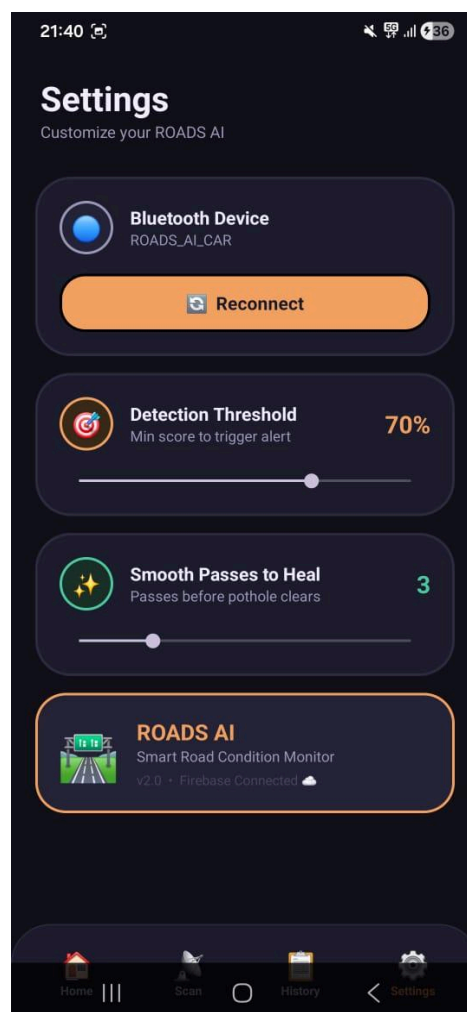


Figure 3.2.4: Settings Screen

Figure 3.2.4 shows the Settings Screen of the R.O.A.D.S Android application. The settings screen provides configuration options for the detection system. A Bluetooth Device card at the top displays the name of the paired hardware device and provides a Reconnect button to manually trigger a new BLE scan. A Detection Threshold

seekbar allows the user to adjust the minimum pothole confidence score required to trigger a detection alert, with the current value displayed as a percentage on the right. A Smooth Passes to Heal seekbar allows the user to configure how many smooth road passes are required before a pothole is automatically marked as repaired in Firebase. At the bottom of the screen, an about card displays the application name ROADS AI, its description as a Smart Road Condition Monitor, the version number, and the Firebase connection status.

CHAPTER 4: TESTING, IMPLEMENTATION AND EVALUATION

4.1 Testing, Implementation and Evaluation

Testing was conducted to ensure that all features of the R.O.A.D.S system were functioning correctly and providing accurate road condition data to the user. Several types of testing were performed, including:

Unit Testing

Unit testing was conducted to verify that each individual component of the R.O.A.D.S system was functioning correctly in isolation before being integrated with the rest of the system.

For the Edge AI model, the dataset was first prepared on Edge Impulse as shown in Figure 4.1. A total of 60 training samples and 3 test samples were collected with a 95%/5% train/test split. The dataset consists of three classes — Bumpy, Potholes, and Smooth — with 20 samples each, ensuring balanced representation across all classes. Each sample was recorded at 77Hz capturing accelerometer data from the MPU6050 sensor mounted on the prototype vehicle chassis.

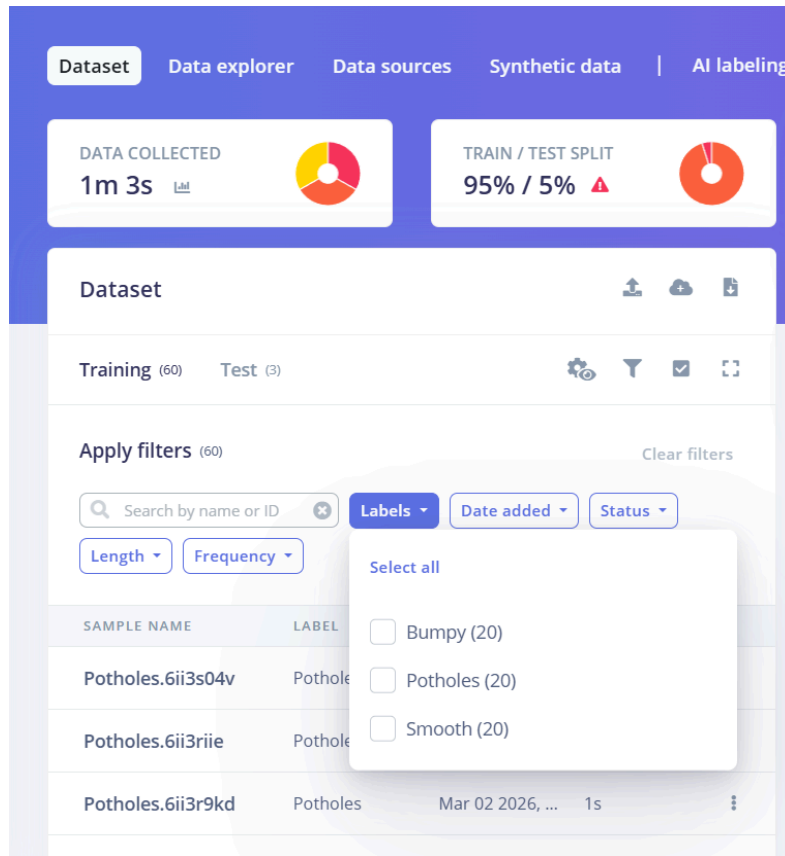


Figure 4.1: Edge Impulse Dataset Collection

The raw vibration data collected from the MPU6050 was processed using a DSP (Digital Signal Processing) block in Edge Impulse. As shown in Figure 4.2, the raw data graph displays the three axes of accelerometer data — accX, accY, and accZ — captured over approximately 1 second for a Potholes sample. The accZ axis shows distinct high-magnitude impulse spikes which represent the physical impact of the vehicle hitting a pothole, clearly differentiating it from smooth road data. The DSP result after filtering further highlights these impulse characteristics, allowing the neural network to extract meaningful features for classification.

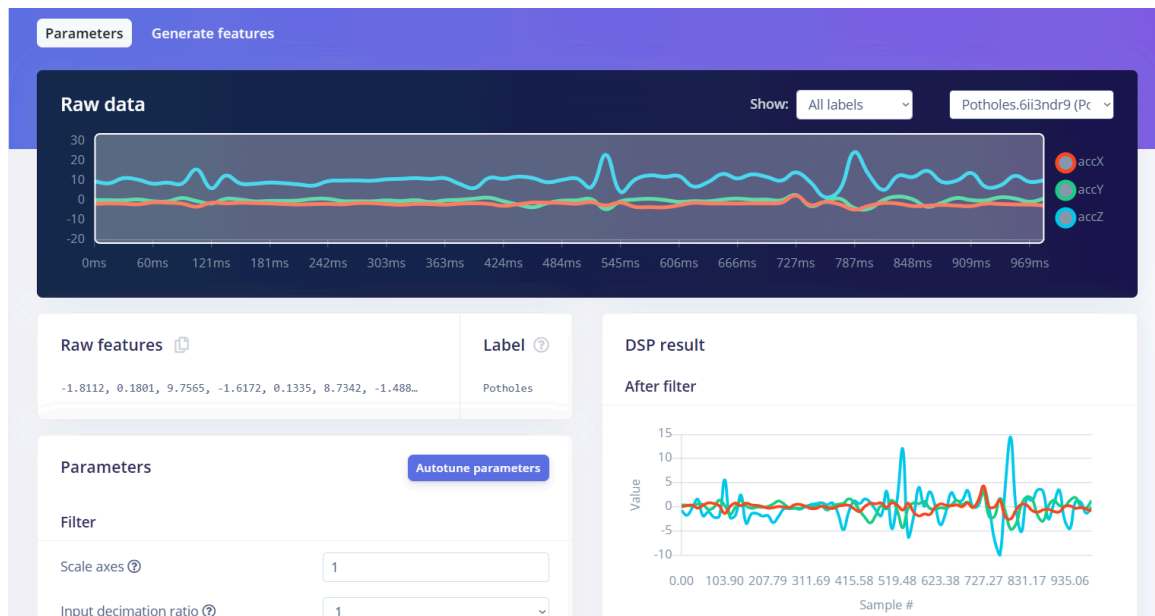


Figure 4.2: Raw Data and DSP Result for Potholes Sample

The neural network architecture used for the Edge AI model is shown in Figure 4.3. The model consists of an Input Layer with 39 features extracted from the spectral analysis processing block, followed by two Dense layers with 20 neurons and 10 neurons respectively, and an Output Layer with 3 classes representing Bumpy, Potholes, and Smooth Road. This lightweight architecture was selected to ensure the model could run efficiently in real time on the ESP32-S3 microcontroller without exceeding its memory and processing constraints.

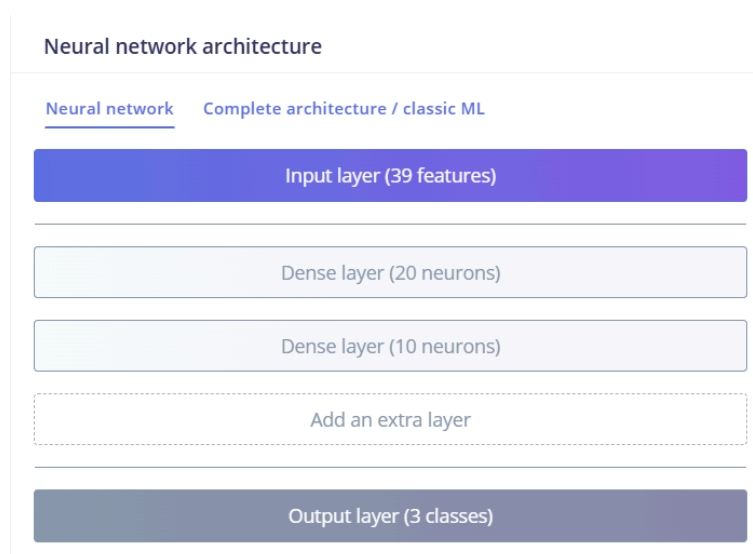


Figure 4.3: Neural Network Architecture

The trained model was evaluated on the validation dataset as shown in Figure 4.4. The model achieved a training accuracy of 100% with a loss value of 0.22 using the Quantized (int8) model version. The confusion matrix shows that all three classes — Bumpy, Potholes, and Smooth — were classified with 100% accuracy and zero misclassification between any classes. The F1 Score for all three classes is 1.00. Additional performance metrics including Area under ROC Curve, Weighted Average Precision, Weighted Average Recall, and Weighted Average F1 Score all recorded a perfect value of 1.00, confirming that the model performs with high reliability before deployment onto the ESP32-S3.

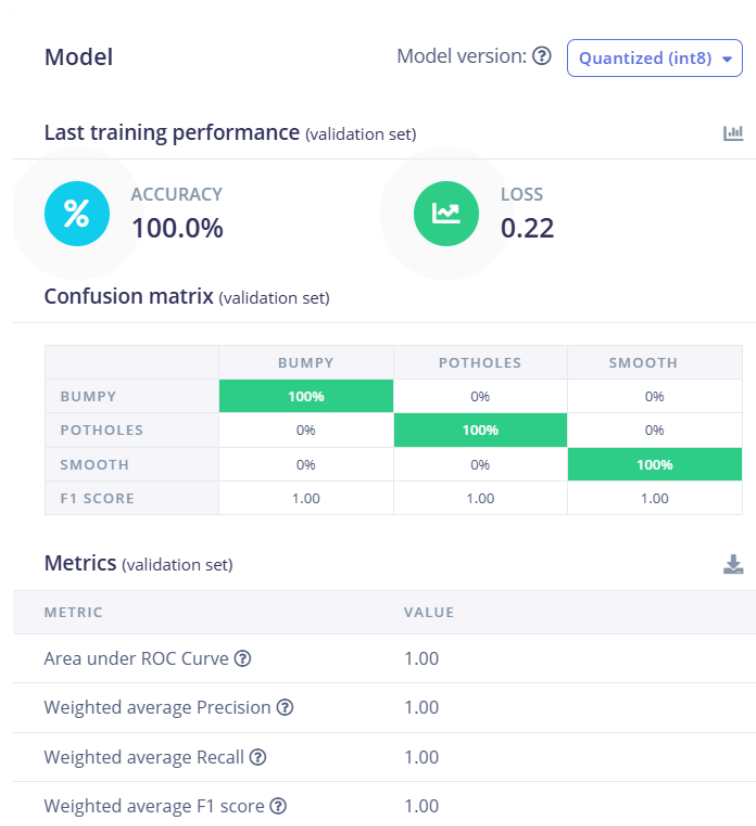


Figure 4.4: Edge Impulse Model Training Result and Confusion Matrix

On the Android application side, individual functions were also tested in isolation including BLE data parsing, Firebase write and read operations, pothole severity calculation, filter functionality in the history screen, and the mark as repaired function. Each function was confirmed to produce the correct output before being connected to other components. Issues such as incorrect BLE packet parsing and

Firestore data not updating the history list in real time were identified and fixed during unit testing before full system integration was carried out.

System Testing

System testing was carried out to verify that the R.O.A.D.S system was functioning as intended across all its components. The system was tested by connecting the ESP32-S3 hardware node to the Android application via Bluetooth Low Energy and simulating different road conditions using the prototype vehicle chassis. Key functionalities tested included BLE device scanning and connection, real-time pothole detection and warning sign display, pothole log saving to Firestore, history list retrieval and filtering, marking potholes as repaired, and the pause and resume road view feature. Each module was tested to ensure that the connection between the hardware node, mobile application, and Firestore database was working correctly. Issues such as BLE connection dropping during tab switches and warning signs not loading from Firestore were identified and resolved during this phase.

Phase Testing

Phase testing was conducted after each phase of the system was developed. The hardware sensor module, Edge AI classification model, BLE communication module, Android application, and Firestore integration were tested separately to ensure that each was functioning correctly before being integrated into the overall system. For example, after developing the Edge AI model using Edge Impulse, the Bumpy, Potholes, and Smooth Road classification results were tested individually on the validation dataset to confirm 100% accuracy before the model was deployed onto the ESP32-S3. After developing the Android application, each screen — Home, Scan, History, and Settings — was tested individually to verify correct data display and user interaction. After successful completion of phase testing, all modules were integrated and tested together to ensure smooth end-to-end functionality and data consistency across the hardware node, mobile application, and Firestore database.

CHAPTER 5: CONCLUSION

5.1 Reflection

The development of R.O.A.D.S has been a comprehensive and rewarding experience that required the integration of multiple technical domains including embedded systems hardware, Edge Artificial Intelligence, Bluetooth Low Energy communication, Android mobile application development, and Firebase cloud computing. Throughout this project, I gained valuable hands-on experience in each of these areas and developed a deeper understanding of how they work together to form a complete IoT system.

One of the most significant challenges encountered during development was the road animation on the Android scan screen stopping unexpectedly after a few seconds. This issue occurred because the animation loop was tied to the view lifecycle and would stop whenever the view was detached from the screen during tab switches. After several attempts using different animation approaches including `postDelayed`, `ValueAnimator`, and `Handler`, the issue was finally resolved by implementing a persistent `RoadView` instance that is created once and reused across tab switches, with the animation driven by a background daemon thread that runs independently of the view lifecycle.

Another challenge was the warning signs not appearing correctly on the road view when reopening the application. The Firebase listener was firing multiple times and calling the load function repeatedly, causing duplicate warning signs to appear on the road. This was resolved by implementing a one-time load flag that ensures historical pothole signs are only loaded once per session.

The Edge AI model development using Edge Impulse was one of the more straightforward aspects of the project. A total of 60 training samples were collected across three classes — Bumpy, Potholes, and Smooth — with 20 samples per class. After training the neural network with an input layer of 39 features, two dense layers of 20 and 10 neurons, and an output layer of 3 classes, the model achieved a training accuracy of 100% with an F1 Score of 1.00 for all classes, which exceeded the initial target of 80% accuracy set in the project objectives.

Overall, this project has significantly strengthened my skills in embedded systems programming, machine learning for constrained devices, Android development in Kotlin, and Firebase cloud integration. It has also developed my problem-solving ability and persistence in debugging complex hardware-software integration issues throughout the development process.

5.2 Recommendation

Based on the development experience and the feedback received from the user survey conducted during the requirement analysis phase, the following recommendations are made for the R.O.A.D.S system.

First, a sound alert should be added to the Android application to accompany the visual warning sign when a pothole is detected. This was suggested by respondents during the survey who felt that an audio cue would be more effective during driving conditions where the driver may not be actively looking at the screen.

Second, based on Question 18 of the survey where 47.4% of respondents preferred receiving road condition warnings on an in-car display rather than a smartphone, future development should explore integrating the system with an in-car display or a heads-up display for a safer and more practical user experience.

Third, the current system does not include GPS coordinate logging. Since 84.2% of survey respondents agreed that sharing road hazard data with authorities is very important, adding GPS functionality would significantly increase the value of the cloud data logged in Firebase by allowing authorities to pinpoint exact pothole locations for repair prioritisation.

5.3 Future Improvement

Several technical enhancements are recommended for future iterations of R.O.A.D.S.

First, a dedicated GPS hardware module should be integrated directly into the ESP32-S3 node to automatically append precise latitude and longitude coordinates to each BLE payload. This would enable accurate geographic mapping of all

detected potholes in Firebase and allow the Android application to display pothole locations on a real-time map view.

Second, the Edge AI model should be further improved by collecting more diverse training data across different vehicle types such as motorcycles and heavy trucks, and at varying vehicle speeds. Although the current model achieved 100% accuracy on the validation dataset, real-world performance across diverse conditions may vary and additional data would help improve robustness.

Third, a web-based dashboard should be developed to allow municipal authorities and fleet managers to access all pothole records on a real-time map, generate severity heatmaps, export maintenance reports, and monitor repair progress. This would fully realise the crowd-sourced infrastructure monitoring potential of R.O.A.D.S and address the strong demand identified in the survey where 100% of respondents agreed on the importance of sharing road hazard data with authorities.

Fourth, based on survey feedback suggesting the system should be affordable, a low-cost production version of the hardware node should be designed using a smaller and cheaper microcontroller board to make the system accessible for mass deployment across logistics fleets and municipal vehicles throughout Malaysia.

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APPENDICES

Appendix I: Survey through Google Form

R.O.A.D.S (Real-Time Onboard Anomaly Detection System) – User Survey

This survey is conducted to gather information about road hazard experiences and awareness of road monitoring systems. Your responses will be used for academic research purposes only.

SECTION A: Respondent Background

1. What is your age range? *

- ☐ Below 18
- ☐ 18 – 25
- ☐ 26 – 35
- ☐ 36 – 45
- ☐ Above 45

2. What is your occupation? *

- ☐ Student
- ☐ Working adult
- ☐ Self-employed
- ☐ Others

3. How often do you travel by road? *

- ☐ Daily
- ☐ A few times a week
- ☐ Occasionally
- ☐ Rarely

4. What type of vehicle do you usually use? *

- ☐ Motorcycle
- ☐ Car
- ☐ Public transport
- ☐ Others

SECTION B: Road Hazard Experience

5. How often do you encounter potholes or road damage during your daily travel? *

- ☐ Every day
- ☐ A few times a week
- ☐ Once a week
- ☐ Rarely
- ☐ Never

6. Have you ever experienced vehicle damage caused by potholes or road anomalies? (e.g. tyre puncture, suspension damage) *

☐ Yes

☐ No

7. Have you ever been involved in or witnessed an accident caused by a pothole or road damage? *

☐ Yes

☐ No

8. How would you rate the overall road condition in your area? *

☐ Very poor

☐ Poor

☐ Average

☐ Good

☐ Very good

9. Have you ever reported a road damage or pothole to the authorities? *

☐ Yes

☐ No

10. If yes, how did you report it? *

- ☐ Mobile application (e.g. Waze, MyJPJ)
- ☐ Social media
- ☐ Called the authority directly
- ☐ Never reported
- ☐ Others

SECTION C: Awareness of Road Monitoring Systems

11. Are you aware of any existing automated road condition monitoring system? *

- ☐ Yes
- ☐ No

12. Have you ever used any application to report road damage? *

- ☐ Yes
- ☐ No

13. If yes, were you satisfied with the response from the authorities after reporting? *

- ☐ Very satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Not satisfied
- ☐ Very not satisfied

SECTION D: System Interest and Needs

14. Do you think there is a need for an automated system that can detect and warn drivers about potholes in real time? *

- ☐ Strongly agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

15. Would you be interested in using a system that automatically detects potholes and displays a warning on your phone while driving? *

- ☐ Very interested
- ☐ Interested
- ☐ Neutral
- ☐ Not interested
- ☐ Not interested at all

16. Which feature do you think is most important in a road monitoring system? *
(Choose one)

- ☐ Real-time warning to the driver
- ☐ Automatic cloud logging of pothole locations
- ☐ Repair status tracking
- ☐ Historical data and statistics
- ☐ All of the above

17. How important is it for road hazard data to be shared with local authorities for maintenance purposes? *

- ☐ Very important
- ☐ Important
- ☐ Neutral
- ☐ Not important
- ☐ Not important at all

18. What device would you prefer to receive road condition warnings on? *

- ☐ Smartphone
- ☐ In-car display
- ☐ Smartwatch
- ☐ Others

19. Would you be willing to have this system installed in your vehicle? *

- ☐ Yes, definitely
- ☐ Yes, if it is affordable
- ☐ Maybe
- ☐ No

20. Any suggestions or comments regarding road condition monitoring systems?

Your answer _____

Appendix II: Responses

Timestamp	1. What is your age?	2. What is your gender?	3. How often do you use a car?	4. What is your vehicle type?	5. How often do you use a car?	6. How often do you use a car?	7. How often do you use a car?	8. How much do you use a car?	9. If you have a car, how often do you report it?	10. If you have a car, how often do you report it?	11. How often do you use a car?	12. How often do you use a car?	13. Do you think it is important to have a car?	14. Would you use a car?	15. Which feature do you think is most important?	16. Which feature do you think is most important?	17. How important is it to have a car?	18. What device do you use to report a car?	19. Would you be willing to pay for a car?	20. Any suggestions or comments regarding the car?
3/17/2026 4:12:17	18 - 25	Student	A few times a week	Motorcycle	Every day	Yes	Yes	Average	No	Never reported	No	No	Very not satisfied	Strongly agree	Very interested	All of the above	Very important	In-car display	Yes, definitely	The system should be affordable, accurate, and easy to use.
3/17/2026 6:23:28	18 - 25	Student	Daily	Car	Once a week	Yes	No	Good	No	Never reported	Yes	No	Neutral	Strongly agree	Very interested	All of the above	Very important	Smartphone	Yes, definitely	
3/17/2026 7:04:52	18 - 25	Student	A few times a week	Car	Every day	Yes	No	Average	No	Never reported	No	No	Neutral	Strongly agree	Very interested	All of the above	Very important	Smartphone	Yes, definitely	
3/17/2026 7:55:56	Above 45	Others	Occasionally	Car	Rarely	No	No	Good	No	Called the authority directly	No	No	Satisfied	Strongly agree	Very interested	Rear-view warning to the driver	Very important	In-car display	Yes, if it is affordable	
3/17/2026 8:05:20	18 - 25	Student	Rarely	Car	Every day	Yes	Yes	Average	No	Never reported	No	No	Neutral	Neutral	Interested	Rear-view warning to the driver	Very important	In-car display	Yes, if it is affordable	
3/17/2026 8:11:44	Above 45	Others	Daily	Car	A few times a week	Yes	No	Average	No	Others	Yes	No	Neutral	Agree	Interested	All of the above	Important	Smartphone	Yes, definitely	
3/17/2026 8:21:09	Above 45	Others	A few times a week	Car	Rarely	Yes	No	Very poor	No	Never reported	Yes	No	Not satisfied	Strongly agree	Very interested	All of the above	Very important	Others	Maybe	
3/17/2026 8:23:04	36 - 45	Working adult	Daily	Car	Every day	Yes	No	Average	No	Never reported	No	No	Neutral	Strongly agree	Very interested	All of the above	Very important	In-car display	Yes, definitely	
3/17/2026 8:36:31	Above 45	Working adult	Daily	Car	Every day	No	Yes	Average	No	Never reported	No	No	Not satisfied	Strongly agree	Interested	All of the above	Very important	Smartphone	Yes, if it is affordable	
3/17/2026 9:56:21	26 - 35	Working adult	Daily	Motorcycle	A few times a week	Yes	Yes	Average	No	Never reported	No	No	Neutral	Agree	Interested	All of the above	Very important	In-car display	Yes, if it is affordable	just focus on your journey, please concentrate on the road.
3/17/2026 10:40:15	36 - 45	Self-employed	Daily	Car	Once a week	Yes	Yes	Average	No	Never reported	No	No	Neutral	Strongly agree	Very interested	All of the above	Very important	Smartphone	Yes, if it is affordable	
3/17/2026 12:32:58	Above 45	Others	Daily	Car	A few times a week	Yes	No	Good	No	Never reported	No	No	Neutral	Agree	Interested	All of the above	Very important	Smartphone	Yes, definitely	
3/17/2026 13:50:03	26 - 35	Working adult	Daily	Car	A few times a week	Yes	Yes	Average	No	Never reported	No	No	Neutral	Strongly agree	Very interested	Rear-view warning to the driver	Very important	Smartphone	Yes, definitely	
3/17/2026 16:09:59	26 - 35	Working adult	Daily	Car	Every day	Yes	No	Average	No	Never reported	No	No	Neutral	Strongly agree	Very interested	All of the above	Very important	Smartphone	Yes, if it is affordable	This should be implemented all over Malaysia, the more the better.
3/17/2026 18:59:00	18 - 25	Student	A few times a week	Car	A few times a week	No	Yes	Good	No	Mobile application (e.g. Waze, Mapbox)	Yes	Yes	Neutral	Agree	Interested	Rear-view warning to the driver	Important	In-car display	Yes, definitely	
3/17/2026 19:03:03	18 - 25	Student	A few times a week	Motorcycle	Every day	No	No	Average	No	Never reported	Yes	No	Neutral	Agree	Very interested	All of the above	Very important	In-car display	Yes, definitely	no
3/17/2026 19:18:14	18 - 25	Student	Occasionally	Car	Once a week	Yes	Yes	Average	No	Never reported	Yes	Yes	Satisfied	Agree	Interested	Automatic cloud logging of profile locations	Important	Smartphone	Yes, definitely	hope
3/17/2026 20:04:34	18 - 25	Student	A few times a week	Car	A few times a week	No	Yes	Average	No	Never reported	No	No	Neutral	Strongly agree	Very interested	Rear-view warning to the driver	Very important	In-car display	Yes, definitely	
3/17/2026 20:42:45	18 - 25	Student	A few times a week	Car	Every day	Yes	No	Poor	No	Never reported	No	No	Neutral	Strongly agree	Very interested	Rear-view warning to the driver	Very important	In-car display	Yes, definitely	no
3/17/2026 22:17:00	26 - 35	Working adult	Daily	Car	A few times a week	Yes	No	Average	No	Never reported	No	No	Neutral	Strongly agree	Interested	All of the above	Important	Smartphone	Yes, if it is affordable	